

From The Energy Explorer:

The ECREC Challenge:

“The Energy Classification, Reasoning, Explanation, and Clue-Matching Challenge.”

Game instructions:

Set Up:

1. Maximum 3 team-mates per group
2. Have the “Energy Source cards facing down on the “Energy Source Card panel”
3. Have the clue cards besides the board game, facing up.

Playing the game:

1. Pick a Energy Source Card.
2. Think what Energy Source is described by that card.
3. Classify the Energy Source Card you just picked by placing it in the correct category. Is it a Renewable or Nonrenewable resource?
4. Explain your classification and description by giving a reason for your decision.
5. Look at the Clue cards, read them carefully and decide which clue card is associated with the card you just picked.

Points in the Game:

1 point for:

Thinking- Group gets 1 point if they made the right consideration of where the energy comes from.

Classification- Group gets 1 mark for placing the Energy Source Card in the correct category.

Explanation- Group gets final 1 point for giving a correct, valid, logical reason for their decision.

Clue matching- Group gets 1 point for matching clue card to the energy source card

Total point per complete cycle = 4 point.

THE ECREC CHALLENGE

Energy Classification, Reasoning, Explanation & Clue-Matching Challenge

IDENTIFY • THINK • CLASSIFY • EXPLAIN • MATCH

ENERGY SOURCES

CLUE CARDS

Can you discover the energy source, classify it correctly, explain your reasoning, and match the right clue?

RENEWABLE ENERGY

CARD HOLDER PANEL

NON-RENEWABLE ENERGY

CARD HOLDER PANEL

ENERGY SOURCES

CARD HOLDER PANEL

ENERGY SOURCE 01

SOLAR ENERGY

Can you classify me?

ENERGY TYPE & WATER

ENERGY SOURCE 02

WIND ENERGY

Can you classify me?

ENERGY TYPE & WATER

ENERGY SOURCE 03

HYDROELECTRIC ENERGY

Can you classify me?

ENERGY TYPE & WATER

ENERGY SOURCE 04

BIOMASS ENERGY

Can you classify me?

ENERGY TYPE & WATER

ENERGY SOURCE 05

COAL

Can you classify me?

ENERGY TYPE & WATER

ENERGY SOURCE 06

OIL

Can you classify me?

ENERGY TYPE & WATER

ENERGY SOURCE 07

NATURAL GAS

Can you classify me?

ENERGY TYPE & WATER

CLUE CARD 01

I come from a natural source that provides light and heat to Earth. What energy source am I?

WHAT AM I?

TYPE • TYPE • EXPLAIN

CLUE CARD 02

I use the movement of air to turn turbines and generate electricity. What energy source am I?

WHAT AM I?

TYPE • TYPE • EXPLAIN

CLUE CARD 03

I use moving water to turn turbines and generate electricity. What energy source am I?

WHAT AM I?

TYPE • TYPE • EXPLAIN

CLUE CARD 04

I can be produced from organic material such as plants and animal waste. What energy source am I?

WHAT AM I?

TYPE • TYPE • EXPLAIN

CLUE CARD 05

I am a fossil fuel formed over millions of years. Once used, I cannot be replaced quickly. What energy source am I?

WHAT AM I?

TYPE • TYPE • EXPLAIN

CLUE CARD 06

I am a fossil fuel formed underground and can be processed into fuels used for transport. What energy source am I?

WHAT AM I?

TYPE • TYPE • EXPLAIN

CLUE CARD 07

I am a fossil fuel found underground and can be burned to produce heat and electricity. What energy source am I?

WHAT AM I?

TYPE • TYPE • EXPLAIN



THE ECREC CHALLENGE

Energy Classification, Reasoning, Explanation & Clue-Matching Challenge

IDENTIFY • THINK • CLASSIFY • EXPLAIN • MATCH



ENERGY SOURCES



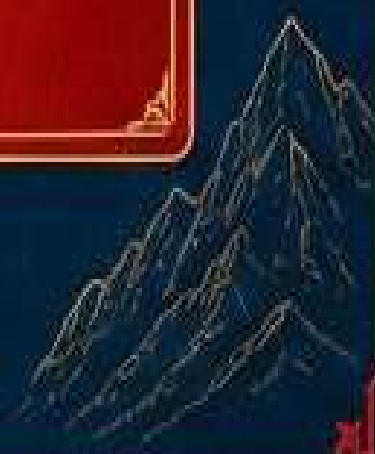
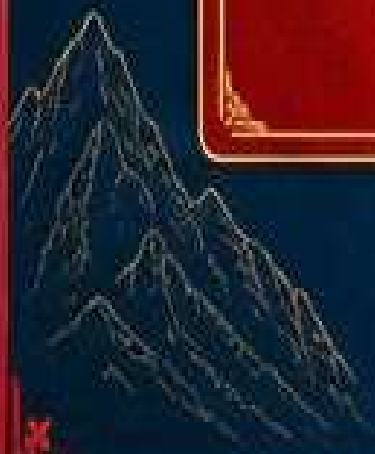
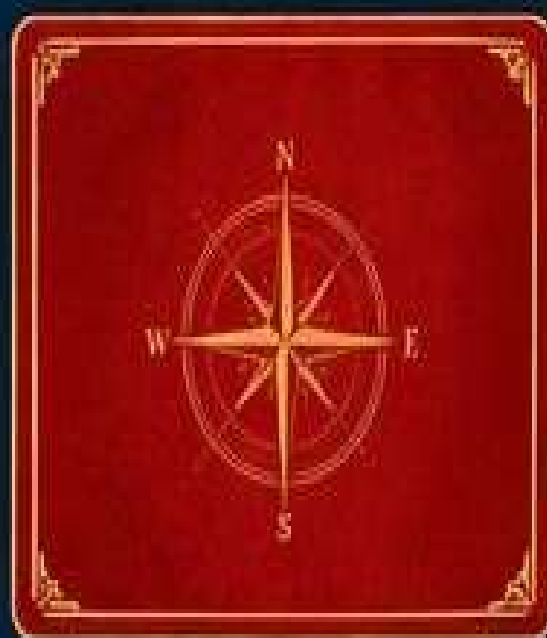
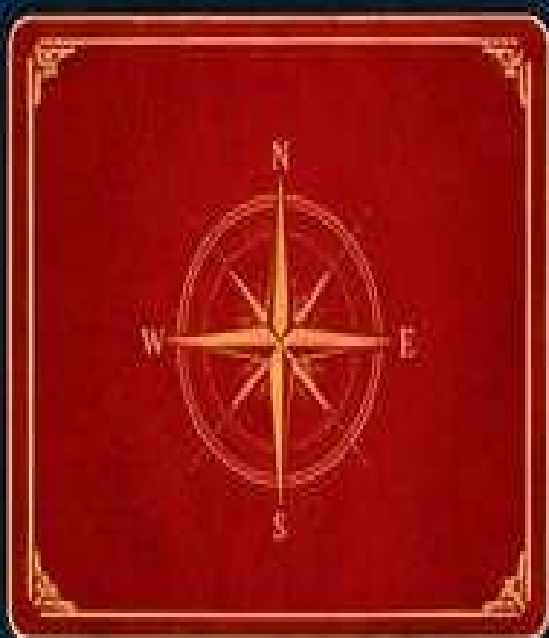
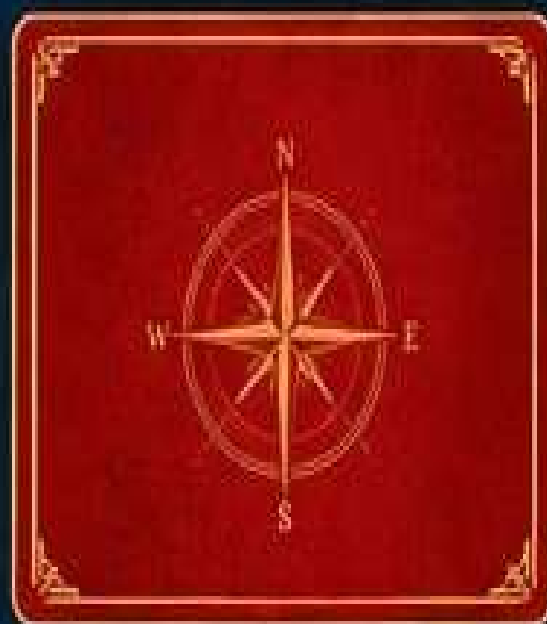
CLUE CARDS

Can you discover the energy source, classify it correctly, explain your reasoning,
and match the right clue?



RENEWABLE ENERGY

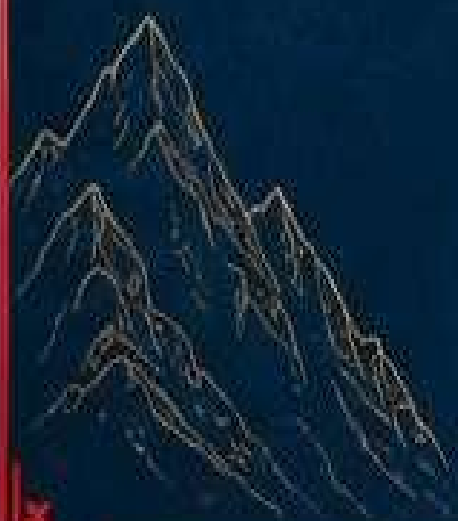
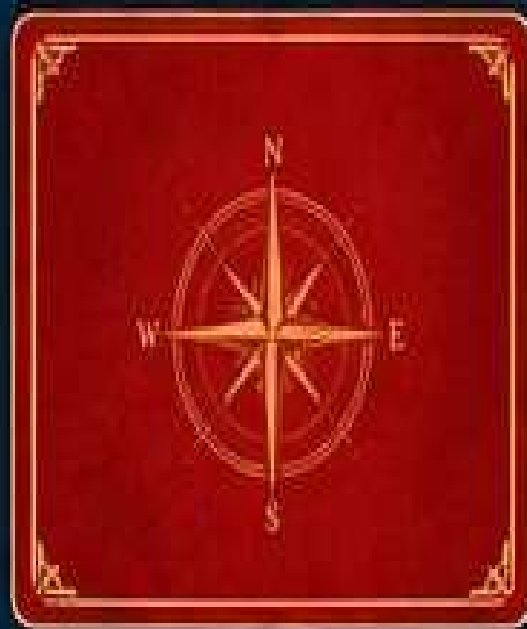
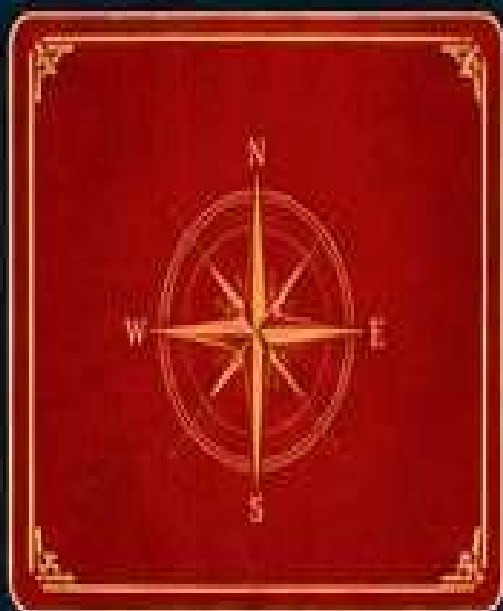
CARD HOLDER PANEL





NON-RENEWABLE ENERGY

CARD HOLDER PANEL



ENERGY SOURCES

CARD HOLDER PANEL



ENERGY SOURCE

01



SOLAR ENERGY

Can you classify me?

ENERGY SORT & MATCH

ENERGY SOURCE

02



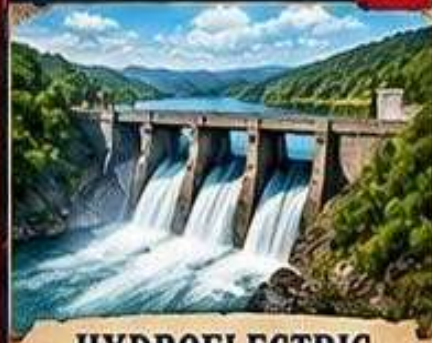
WIND ENERGY

Can you classify me?

ENERGY SORT & MATCH

ENERGY SOURCE

03



HYDROELECTRIC ENERGY

Can you classify me?

ENERGY SORT & MATCH

ENERGY SOURCE

04



BIOMASS ENERGY

Can you classify me?

ENERGY SORT & MATCH

ENERGY SOURCE

05



COAL

Can you classify me?

ENERGY SORT & MATCH

ENERGY SOURCE

06



OIL

Can you classify me?

ENERGY SORT & MATCH

ENERGY SOURCE

07



NATURAL GAS

Can you classify me?

ENERGY SORT & MATCH



CLUE CARD

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CLUE CARD

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FIND • SORT • EXPLAIN

CLUE CARD

07

I am a fossil fuel found underground and can be burned to produce heat and electricity.
What energy source am I?

WHAT AM I?

FIND • SORT • EXPLAIN



For the purpose of making it easy with the design process, the Energy Source Cards and the Clue Cards have corresponding so you are able to understand how the Energy Sources and Clue Cards are linked. Basically, consider it as a visual rubric.

Once the game has been designed, cut out these circles and cover only the Clue Cards as shown in the diagram below.

Print out the cover ups attached on the side of this paper

Before



After



Reference list.

OpenAI. 2026. *Generate Board Game images*. [online] Available at: < [Design Card Holder Panels](#) > [Accessed 10 September 2026].

Zondo, N., T. 2026. *Energy Sort & Match Challenge Game instructions and Printables*. [online] Available at: < [Energy Explorer | Grade 7 Natural Sciences](#) > [Accessed 10 September 2026].

Acknowledgements.

This physical resource, deriving from the digital learning resource, Energy Explorer | Grade 7 Natural Sciences, was designed and developed by Njabulo Zondo as an educational resource to support Grade 7 learners' understanding of renewable and non-renewable energy sources.

The content, activities, educational game, worksheets, and overall structure were developed and adapted by the author for educational purposes, with consideration of the South African Curriculum and Assessment Policy Statement (CAPS) requirements for Natural Sciences.

Various external educational resources, images, videos, and digital materials have been used where appropriate. These resources have been acknowledged and referenced accordingly, and their original creators and sources remain credited.

ChatGPT (GPT-5.6 Luna), developed by OpenAI, was used as an AI-assisted co-design and writing support tool during the development of this resource. Its assistance included brainstorming, refining educational explanations, improving wording and structure, troubleshooting aspects of the website code, and supporting the development of selected learning activities and digital-resource ideas. The final selection, adaptation, editing, educational decisions, design, implementation, and responsibility for the completed resource remain with Njabulo Zondo.

This acknowledgement is included to promote transparency regarding the development process and the use of digital and AI-assisted tools in the creation of the resource.